The Minimum Number of Triangles in an $(n, 5, 3)$-Graph *

Yanan Hu[1], Chengli Li[2], Feng Liu[3] and Leilei Zhang[4,5†]

Abstract

An (n, k, t)-graph is a graph on n vertices in which every set of k vertices contains a copy of K_t. Turán's theorem implies that an $(n, k, 2)$-graph with the minimum number of edges is an equitable disjoint union of cliques. Recently, Baumann and Briggs (*Electron. J. Combin.*, 2025) settled the Strong (n, k, t)-Conjecture by determining the minimum number of edges in an (n, k, t)-graph. In their paper, they proposed the further problem of determining the minimum number of s-cliques in an (n, k, t)-graph. The corresponding problem for $(n, k, 2)$-graphs is a classical problem of Erdős and has been extensively studied. The case of $(n, 4, 3)$-graphs has also been determined. Thus, Baumann and Briggs pointed out that the smallest interesting open problem is to determine the minimum number of triangles in an $(n, 5, 3)$-graph. In this note, we resolve this problem and characterize all extremal graphs.

Keywords. (n, k, t)-graph; triangle; Goodman's identity.

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1 Introduction

We consider finite simple graphs and use standard terminology and notation from [3]. For a graph G, we write $V(G)$ and $E(G)$ for its vertex set and edge set, respectively. If $S \subseteq V(G)$, then $G[S]$ denotes the subgraph of G induced by S, and we write $G - S := G[V(G) \setminus S]$. For a vertex $v \in V(G)$, we denote its neighbourhood and degree by $N_G(v)$ and $d_G(v)$; we omit the subscript when the underlying graph is clear. The complement of G is denoted by $\overline{G}$.

An *empty graph* is a simple graph in which no two vertices are adjacent. A graph is *triangle-free* if it contains no copy of K_3. A cycle is *induced* if the subgraph on its vertex set has no edge other than the edges of the cycle. A graph is *bipartite* if its vertex set can be partitioned into two independent sets. A *matching* is a set of edges no two of which share an endpoint. A set of vertices is *independent* if no two of its vertices are adjacent.

For integers $n \geq k \geq t \geq 2$, we call a graph G an (n, k, t)*-graph* if G is an n-vertex graph in which every k-vertex subset contains a clique of order t. The case $t = 2$ is a reformulation of a classical extremal problem. An $(n, k, 2)$-graph has no independent set of order k, and Turán's theorem [11] implies that the minimum number of edges is attained by the equitable disjoint union of $k - 1$ cliques. Hoffman, Johnson, and McDonald [7] initiated the systematic study of edge minimization in general (n, k, t)-graphs. They conjectured that every (n, k, t)-graph attaining the minimum number of edges is a disjoint union of cliques. Noble, Johnson,

* *Email addresses*: mathdzhang@163.com (L. Zhang).

† Corresponding author.

Hoffman, and McDonald [9] subsequently proved this conjecture for $k \le 2t-2$. Baumann and Briggs [1] subsequently proved it for all admissible triples (n,k,t).

This naturally leads to the corresponding extremal problem for cliques of larger order. For $s \ge 2$, let $N_s(G)$ denote the number of copies of K_s in G, and define

$$\mu_s(n,k,t) := \min\bigl\{N_s(G) : G \text{ is an } (n,k,t)\text{-graph}\bigr\}.$$

When $t=2$, this question belongs to the line of work on minimizing the number of cliques in graphs with bounded independence number, initiated by Erdős [5]; see also [4, 8, 10, 12, 13]. For general t, Baumann and Briggs asked for the determination of $\mu_s(n,k,t)$. When $k \le 2t-2$, the problem is settled by a theorem of Noble, Johnson, Hoffman and McDonald [9], which states that every (n,k,t)-graph contains a clique of order $n-k+t$. Together with the construction $(k-t)K_1 \cup K_{n-k+t}$, this yields

$$\mu_s(n,k,t) = \binom{n-k+t}{s}.$$

Here $(k-t)K_1 \cup K_{n-k+t}$ denotes the disjoint union of $k-t$ isolated vertices and a complete graph of order $n-k+t$.

Therefore, the smallest case not covered by this result is $(s,k,t)=(3,5,3)$. Accordingly, Baumann and Briggs [1] posed the following problem.

Problem 1.1 (Baumann and Briggs [1])**.** For $n \ge 5$, determine the minimum possible number of triangles in an $(n,5,3)$-graph.

Recently, Bodnár and Pikhurko established the asymptotic formula

$$\mu_3(n,5,3) = \left(\frac{1}{4} + o(1)\right)\binom{n}{3}$$

together with a stability result. Their more general theorem on $(n,2t-1,t)$-graphs also determines the exact value and the extremal graphs for all sufficiently large n [2]. In the case $(s,k,t)=(3,5,3)$, their extremal graphs consist of two disjoint balanced cliques together with an arbitrary matching between the two clique parts. We prove that the same formula holds for $n \ge 5$ and that this extremal description remains valid for $n \ge 6$. The case $n=5$ is exceptional: the extremal graphs are precisely those containing exactly one triangle. We now state our main result. A matching is allowed to be empty.

Theorem 1.2. *Let $n \ge 5$. Then*

$$\mu_3(n,5,3) = \binom{\lfloor n/2 \rfloor}{3} + \binom{\lceil n/2 \rceil}{3}. \tag{1}$$

When $n=5$, the extremal graphs contain exactly one triangle. For $n \ge 6$, the extremal graphs are precisely those obtained from the vertex-disjoint union $K_{\lfloor n/2 \rfloor} \cup K_{\lceil n/2 \rceil}$ by adding an arbitrary matching between the two clique parts.

For reference, the value in (1) can be written without floor and ceiling signs as

$$\mu_3(n,5,3) = \begin{cases} \dfrac{n(n-2)(n-4)}{24}, & n \text{ even}, \\[2ex] \dfrac{(n-1)(n-2)(n-3)}{24}, & n \text{ odd}. \end{cases}$$

In particular,

$$\lim_{n\to\infty} \frac{\mu_3(n,5,3)}{\binom{n}{3}} = \frac{1}{4}.$$

The remainder of the paper is organized as follows. In Section 2, we fix notation and collect the counting and structural lemmas used in the proof. In Section 3, we prove Theorem 1.2, treating even and odd orders separately.

2 Preliminaries

We write K_r, C_r, and $K_{a,b}$ for the complete graph of order r, the cycle of length r, and the complete bipartite graph with parts of orders a and b, respectively. For a finite set X and a nonnegative integer r, let $\binom{X}{r}$ denote the family of all r-element subsets of X. We adopt the convention that $\binom{q}{r} = 0$ whenever $0 \le q < r$.

For a graph G and an integer $r \ge 1$, we recall that $N_r(G)$ is the number of copies of K_r in G; in particular, $N_3(G)$ is the number of triangles in G. Similarly, we define $i_r(G)$ to be the number of independent sets of order r. The independence number $\alpha(G)$ is the maximum order of an independent set in G.

We now reformulate the $(n,5,3)$ condition in terms of the complement. For a graph H of order at least five, consider the property

$$\alpha(H[X]) \ge 3 \qquad \text{for every } X \in \binom{V(H)}{5}. \tag{$\mathcal{P}$}$$

Equivalently, every set of five vertices of H contains an independent set of order three.

Proposition 2.1. *Let G be a graph of order $n \ge 5$, and let $H = \overline{G}$. Then G is an $(n,5,3)$-graph if and only if H satisfies ($\mathcal{P}$). Furthermore,*

$$N_3(G) = i_3(H). \tag{2}$$

Proof. For every vertex set $S \in \binom{V(G)}{3}$, we have

$$G[S] \cong K_3 \quad \Longleftrightarrow \quad E(H[S]) = \varnothing.$$

Consequently, the triangles of G are in one-to-one correspondence with the independent sets of order three in H, proving (2).

Now let $X \in \binom{V(G)}{5}$. The induced subgraph $G[X]$ contains a triangle if and only if $H[X]$ contains an independent set of order three. Hence every five-vertex subset of G spans a triangle if and only if H satisfies ($\mathcal{P}$), as required. □

The following identity is the graph-theoretic form of Goodman's classical counting formula [6]. For completeness, we include its short proof.

Lemma 2.2 (Goodman [6])**.** *Let Q be a graph on n vertices. Then*

$$N_3(Q) + i_3(Q) = \binom{n}{3} - \frac{1}{2} \sum_{v \in V(Q)} d_Q(v)\big(n - 1 - d_Q(v)\big). \tag{3}$$

Proof. We call a three-element subset of $V(Q)$ homogeneous if it induces either a triangle or an independent set. Let $M(Q)$ denote the number of non-homogeneous triples. Thus, $M(Q)$ counts the three-element subsets inducing exactly one or two edges, and hence

$$\binom{n}{3} = N_3(Q) + i_3(Q) + M(Q). \tag{4}$$

For a fixed vertex $v \in V(Q)$, the quantity

$$d_Q(v)\big(n-1-d_Q(v)\big)$$

counts the three-element subsets $S \subseteq V(Q)$ containing v for which v has exactly one neighbor in $S \setminus \{v\}$. Consequently, the sum

$$\sum_{v \in V(Q)} d_Q(v)\big(n-1-d_Q(v)\big)$$

counts the pairs (S, v) such that $|S| = 3$, $v \in S$, and v has exactly one neighbor in $S \setminus \{v\}$.

If $Q[S]$ has exactly one edge, then precisely the two endpoints of that edge have exactly one neighbor in S. If $Q[S]$ has exactly two edges, then $Q[S]$ is a path on three vertices, and precisely its two endvertices have exactly one neighbor in S. By contrast, no vertex of a triangle or an independent triple has exactly one neighbor within the triple. It follows that every non-homogeneous triple is counted exactly twice, whereas every homogeneous triple is counted zero times. Therefore,

$$\sum_{v \in V(Q)} d_Q(v)\big(n-1-d_Q(v)\big) = 2M(Q).$$

Substituting this equality into (4) yields (3). □

We also record the elementary balancing inequality that will determine the sizes of the two parts in an extremal graph.

Lemma 2.3. *Let $n \geq 3$ be an integer, and let a and b be nonnegative integers with $a + b = n$. Then*

$$\binom{a}{3} + \binom{b}{3} \geq \binom{\lfloor n/2 \rfloor}{3} + \binom{\lceil n/2 \rceil}{3}. \tag{5}$$

Equality holds if and only if $|a - b| \leq 1$.

Proof. Suppose, without loss of generality, that $a \leq b$. If $b - a \geq 2$, we replace (a, b) by $(a+1, b-1)$. This operation preserves the sum $a + b$, while Pascal's identity gives

$$\binom{a+1}{3} + \binom{b-1}{3} - \binom{a}{3} - \binom{b}{3} = \binom{a}{2} - \binom{b-1}{2} < 0.$$

Thus transferring one unit from the larger parameter to the smaller one strictly decreases whenever their difference is at least two.

Repeating this operation eventually produces a pair whose entries differ by at most one. Since their sum remains equal to n, the resulting pair is

$$\left(\left\lfloor \frac{n}{2} \right\rfloor, \left\lceil \frac{n}{2} \right\rceil\right).$$

This proves (5). Moreover, every balancing step is strict, so equality holds if and only if the original pair already differs by at most one, equivalently,

$$\{a, b\} = \left\{\left\lfloor \frac{n}{2} \right\rfloor, \left\lceil \frac{n}{2} \right\rceil\right\}.$$

□

The following lemma treats the case in which H contains a triangle.

Lemma 2.4. *Let H be an n-vertex graph satisfying ($\mathcal{P}$). If H contains a triangle, then*

$$i_3(H) \geq \binom{n-2}{3}.$$

More precisely, if T is the vertex set of a triangle and $W = V(H) \setminus T$, then W is independent, and every pair of vertices in W has a common non-neighbour in T.

Proof. Let $T = \{a, b, c\}$, and take distinct vertices $x, y \in W$. By ($\mathcal{P}$), the five-element set $T \cup \{x, y\}$ contains an independent triple. Such a triple contains at most one vertex of T, because $H[T]$ is complete. It must therefore have the form $\{x, y, z\}$ for some $z \in T$. In particular, $xy \notin E(H)$, and z is a common non-neighbour of x and y.

It follows that all $\binom{|W|}{3}$ triples contained in W are independent. For each pair in W, we also obtain at least one independent triple containing that pair and one vertex of T. These two collections are disjoint, so

$$i_3(H) \geq \binom{|W|}{3} + \binom{|W|}{2} = \binom{n-3}{3} + \binom{n-3}{2} = \binom{n-2}{3}.$$

□

Lemma 2.5. *If a triangle-free graph H satisfies ($\mathcal{P}$), then H is C_5-free.*

Proof. Suppose that H contains a copy of C_5. Since H is triangle-free, this cycle has no chord and is therefore induced. Its five vertices have independence number two, contradicting ($\mathcal{P}$). □

Lemma 2.6. *Let Q be a non-bipartite graph of order n containing neither C_3 nor C_5, and let C be a shortest odd cycle in Q. If $\ell = |V(C)|$, then the following statements hold:*

1. *$\ell \geq 7$, and C is induced;*
2. *every vertex in $V(Q) \setminus V(C)$ has at most two neighbours on C;*
3. $$\sum_{v \in V(C)} d_Q(v) \leq 2n.$$

Proof. The absence of C_3 and C_5 gives $\ell \geq 7$. A chord of C would divide it into two shorter cycles, one of which would be odd. Thus the minimality of C implies that C is induced.

Suppose that a vertex $x \notin V(C)$ has at least three neighbours on C. In their cyclic order, these neighbours divide C into at least three arcs. Each arc has length at least two, since Q is triangle-free. The sum of the arc lengths is odd, so at least one arc has odd length; denote its length by r. The arc together with the two edges joining its endpoints to x forms an odd cycle of length $r + 2$. By the choice of C, we have $r + 2 \geq \ell$, and hence $r \geq \ell - 2$. The remaining arcs have total length at most two, but there are at least two of them and each has length at least two. This contradiction proves the second statement.

Since C is induced, its edges contribute 2ℓ to the sum of the degrees of its vertices. The second statement shows that the vertices outside C contribute at most $2(n-\ell)$ further incidences. Hence

$$\sum_{v \in V(C)} d_Q(v) \leq 2\ell + 2(n - \ell) = 2n,$$

as required. □

3 Proof of the main result

Proof of Theorem 1.2. We first dispose of the case $n = 5$. The $(5, 5, 3)$ condition simply says that G contains a triangle. Consequently,

$$N_3(G) \geq 1 = \binom{2}{3} + \binom{3}{3},$$

and equality holds exactly when G contains one triangle.

We next assume that $n \geq 6$. Let $H = \overline{G}$. By Proposition 2.1, the graph H satisfies $(\mathcal{P})$, and

$$N_3(G) = i_3(H).$$

We consider separately the cases in which H contains a triangle and in which it is triangle-free.

Case 1: H contains a triangle.

Lemma 2.4 gives

$$i_3(H) \geq \binom{n-2}{3}.$$

We compare this bound with the claimed value. If $n = 2m \geq 6$, then

$$\binom{2m-2}{3} - 2\binom{m}{3} = (m-1)(m-2)^2 > 0.$$

If $n = 2m + 1 \geq 7$, then

$$\binom{2m-1}{3} - \binom{m}{3} - \binom{m+1}{3} = \frac{(2m-1)(m-1)(m-2)}{2} > 0.$$

Thus the desired lower bound is strict whenever H contains a triangle. In particular, the complement of an extremal graph of order at least six must be triangle-free.

Case 2: H is triangle-free.

Lemma 2.5 shows that H is also C_5-free. Applying Goodman's identity and using $N_3(H) = 0$, we obtain

$$i_3(H) = \binom{n}{3} - \frac{1}{2} \sum_{v \in V(H)} d(v)\big(n - 1 - d(v)\big), \tag{6}$$

where $d(v) = d_H(v)$.

Subcase 2.1: $n = 2m$ is even.

For every integer d, we have

$$d(2m - 1 - d) \leq m(m-1), \tag{7}$$

with equality if and only if $d \in \{m-1, m\}$. It follows from (6) that

$$\begin{aligned} i_3(H) &\geq \binom{2m}{3} - \frac{1}{2}(2m)m(m-1) \\ &= \binom{2m}{3} - m^2(m-1) \\ &= 2\binom{m}{3}. \end{aligned}$$

This proves the lower bound when n is even.

Suppose that equality holds. Equality must then hold in (7) at every vertex, so

$$d(v) \in \{m-1, m\} \qquad \text{for } v \in V(H). \tag{8}$$

We claim that H is bipartite. Suppose otherwise. Since H contains neither C_3 nor C_5, we may apply Lemma 2.6 to a shortest odd cycle C of length $\ell \geq 7$. The lemma gives

$$\sum_{v \in V(C)} d(v) \leq 4m. \tag{9}$$

On the other hand, (8) gives

$$\sum_{v \in V(C)} d(v) \geq \ell(m-1) \geq 7(m-1).$$

The existence of C implies $2m \geq \ell \geq 7$, and hence $m \geq 4$. But $7(m-1) > 4m$ for $m \geq 4$, contradicting (9). Therefore H is bipartite.

Let (A, B) be a bipartition of H, and write

$$|A| = a \leq b = |B|, \qquad a + b = 2m.$$

Every vertex of B has degree at most a and, by (8), degree at least $m-1$. Hence $a \geq m-1$. Since $a \leq m$, we have $a \in \{m-1, m\}$. If $a = m-1$, then every vertex of B is adjacent to every vertex of A. Each vertex of A would consequently have degree $|B| = m+1$, contrary to (8). We conclude that

$$|A| = |B| = m.$$

Let

$$R = E(K_{m,m}) \setminus E(H)$$

be the set of missing cross-edges with respect to the bipartition (A, B). By (8), every vertex is incident with at most one edge of R. Thus R is a matching and

$$H = K_{m,m} - R.$$

Taking complements, we see that $G[A]$ and $G[B]$ are complete and that the edges of G between A and B form the matching R.

Subcase 2.2: $n = 2m+1$ **is odd.**

Since $n = 2m+1$, we have $n - 1 = 2m$. Moreover, for $v \in V(H)$,

$$d(v)\big(2m - d(v)\big) = m^2 - (d(v) - m)^2.$$

As $|V(H)| = 2m+1$, it follows from (6) that

$$\begin{aligned} i_3(H) &= \binom{2m+1}{3} - \frac{1}{2} \sum_{v \in V(H)} \big(m^2 - (d(v) - m)^2\big) \\ &= \binom{2m+1}{3} - \frac{(2m+1)m^2}{2} + \frac{1}{2} \sum_{v \in V(H)} (d(v) - m)^2. \end{aligned}$$

A direct calculation gives

$$\binom{2m+1}{3} - \frac{(2m+1)m^2}{2} = \binom{m}{3} + \binom{m+1}{3} - \frac{m}{2}.$$

Consequently,

$$i_3(H) = \binom{m}{3} + \binom{m+1}{3} + \frac{1}{2}\left(\sum_{v\in V(H)} (d(v)-m)^2 - m\right). \tag{10}$$

We first suppose that H is not bipartite. Let C be a shortest odd cycle in H, and write $\ell = |V(C)|$. Lemma 2.6 gives $\ell \geq 7$ and

$$\sum_{v\in V(C)} d(v) \leq 2(2m+1) = 4m+2.$$

Consequently,

$$\sum_{v\in V(C)} (m - d(v)) = \ell m - \sum_{v\in V(C)} d(v) \geq (\ell-4)m - 2.$$

The right-hand side is positive because $\ell \geq 7$ and $m \geq 3$. By the Cauchy–Schwarz inequality,

$$\begin{aligned}
\sum_{v\in V(H)} (d(v)-m)^2 &\geq \sum_{v\in V(C)} (d(v)-m)^2 \\
&\geq \frac{\left(\sum_{v\in V(C)}(m-d(v))\right)^2}{\ell} \\
&\geq \frac{\big((\ell-4)m-2\big)^2}{\ell} \\
&\geq \frac{(3m-2)^2}{2m+1}.
\end{aligned}$$

In the last step, we used $7 \leq \ell \leq 2m+1$. Moreover, for $m \geq 3$,

$$(3m-2)^2 - m(2m+1) = 7m^2 - 13m + 4 > 0.$$

Hence

$$\sum_{v\in V(H)} (d(v)-m)^2 > m.$$

Equation (10) now yields

$$i_3(H) > \binom{m}{3} + \binom{m+1}{3}.$$

Thus a non-bipartite graph cannot attain equality.

It remains to consider the case in which H is bipartite. Let (A, B) be a bipartition, and put $|A| = a$ and $|B| = b$. Every triple contained entirely in one part is independent, so

$$i_3(H) \geq \binom{a}{3} + \binom{b}{3}.$$

Since $a + b = 2m+1$, Lemma 2.3 gives

$$i_3(H) \geq \binom{m}{3} + \binom{m+1}{3}.$$

Suppose that equality holds. The equality statement in Lemma 2.3 forces the part sizes to be m and $m+1$. Moreover, every independent triple must lie entirely in one part. Let

$$R = E(K_{m,m+1}) \setminus E(H)$$

be the graph of missing cross-edges. Suppose that two edges a_1b and a_2b of R share the endpoint b, where $a_1, a_2 \in A$ and $b \in B$. Then none of the three pairs a_1a_2, a_1b, and a_2b is an edge of H, so $\{a_1, a_2, b\}$ is an independent triple meeting both parts. Conversely, every independent triple meeting both parts determines two edges of R with a common endpoint. Hence H has no independent triple meeting both parts if and only if R is a matching. Equality therefore implies that

$$H = K_{m,m+1} - R$$

for some matching R. Taking complements, we obtain two vertex-disjoint cliques of orders m and $m+1$, with the edges between them forming the matching R.

We finish by verifying that every graph described in the theorem is indeed extremal. Let A and B be disjoint sets of orders $\lfloor n/2 \rfloor$ and $\lceil n/2 \rceil$. We place a clique on each of A and B and add an arbitrary matching between them. Every set of five vertices contains at least three vertices from one of A and B, and those three vertices form a triangle. Hence the resulting graph is an $(n, 5, 3)$-graph.

Since the edges between A and B form a matching, no two of them share a common endpoint. Any triangle meeting both A and B would contain two edges between A and B with a common endpoint, which is impossible. Therefore, every triangle lies entirely in A or entirely in B, and hence the number of triangles is exactly

$$\binom{|A|}{3} + \binom{|B|}{3} = \binom{\lfloor n/2 \rfloor}{3} + \binom{\lceil n/2 \rceil}{3}.$$

This proves sharpness and completes the characterization of all equality cases. □

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Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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[1]School of Science, Shanghai Institute of Technology, Shanghai 201418, China
hg@teachermail.sit.edu.cn(Y. Hu)
[2]School of Mathematical Sciences, South China Normal University, Guangzhou 510631, China
lichengli0130@126.com(C. Li)
[3]School of Mathematical Sciences, Shanghai Jiao Tong University, Shanghai 200240, China
liufeng0609@126.com(F. Liu)
[4]School of Mathematics and Statistics, Central China Normal University, Wuhan 430079, China
[5]Faculty of Environment and Information Sciences, Yokohama National University, Yokohama 240-8501, Japan
mathdzhang@163.com(L. Zhang)